

David H. Liu

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WORK EXPERIENCE

Senior Software Engineer, Google DeepMind

2024 - present

GenAI Systems, Auto-Tools

- Productionized a scalable A/B testing system for fleet-scale ML performance engineering, improving TPU duty cycle up to 4x; adopted by 4 teams across 3 organizations.
- Optimized deployment system for XLA autotuning and scaled annual fleetwide TPU savings by 2x.
- Lead compiler optimization for Auto-Tools; established technical roadmaps and expanded tuning capabilities to unlock new compiler knobs.
- Design agentic solutions to increase team's velocity via self-improvement loops and automating high-toil tasks.

Research Intern, Microsoft Research

Summer 2020

Mobility and Networking Group

- Designed benchmarks for serverless computing systems and applications on the Azure Kubernetes Service (AKS).

Software Engineer, Nimble Storage

2015 - 2017

Data Protocol Team

- Led next-gen Fibre Channel kernel driver development, managing cross-company milestones with Broadcom and landing significant throughput improvement.

EDUCATION

Princeton University, Princeton, NJ

2017 - 2023

Ph.D. in Computer Science

Thesis: A Serverless Architecture for Application-Level Orchestration

Advisor: Amit Levy

Duke University, Durham, NC

2011 - 2015

B.S.E. in Electrical and Computer Engineering, minor in Mathematics

SKILLS

Python, Rust, C, C++, Kubernetes, PyTorch, Docker, Linux kernel drivers

PUBLICATIONS

CATWILD: Compiler Autotuning for TPU workloads in the Wild

Ignacio Cano, Yu Wang, Mike Burrows, Ziqiang Feng, David H. Liu, Tengyu Sun, Arissa Wongpanic, Phitchaya Mangpo Phothilimthana, Emma Sevastian, et al.

Ninth Annual Conference on Machine Learning and Systems (MLSys '26), Bellevue, WA, May 2026

Doing More with Less: Orchestrating Serverless Applications without an Orchestrator

David H. Liu, Amit Levy, Shadi Noghabi, Sebastian Burckhardt

Proc. 20th Symposium on Networked Systems Design and Implementation (NSDI '23), Boston, MA, April 2023

How Low Can You Go? Practical cold-start performance limits in FaaS

Yue Tan, David H. Liu, Nanqinqin Li, Amit Levy

ArXiv Technical Report:2109.13319, Sept. 2021

Pyronia: Intra-Process Access Control for IoT Applications

Marcela S. Melara, David H. Liu, Michael J. Freedman

ArXiv Technical Report:1903.01950, March 2019

SandTrap: Tracking Information Flows On Demand with Parallel Permissions

Ali Razeen, Alvin R. Lebeck, David H. Liu, Alexander Meijer, Valentin Pistol, Landon P. Cox

The 16th ACM International Conference on Mobile Systems, Applications, and Services (MobiSys '18), June 2018